

The three hypotheses: BM1, OU1, OUs

sparseou — one gene, one tree, regimes supplied

Setup

A rooted **ultrametric** tree, n tips, height H . One gene's tip values are $\mathbf{y} \in \mathbb{R}^n$. The supplied regime painting gives $r(u) \in \{1, \dots, K\}$, the regime on the branch at time u . Two matrices are fixed by the tree alone:

$$D_{ij} = 2(H - t_{\text{mrca}(i,j)}) \quad (\text{patristic distance}), \quad C_{ij} = t_{\text{mrca}(i,j)} \quad (\text{shared path length}).$$

Write $\mathbf{K}(\alpha)$ for the elementwise exponential

$$[\mathbf{K}(\alpha)]_{ij} = e^{-\alpha D_{ij}},$$

which is *not* a matrix exponential. All three hypotheses are multivariate normal over the tips and differ only in the **mean**.

BM1 — neutral drift

Brownian motion started at θ_{bm} , no pull towards anything; variance accumulates with shared ancestry.

$$\mathbf{y} \sim \mathcal{N}_n(\theta_{\text{bm}} \mathbf{1}_n, \sigma_{\text{bm}}^2 \mathbf{C} + \tau^2 \mathbf{I})$$

Free parameters: θ_{bm} (root state), σ_{bm} (drift rate), τ (measurement error).

OU1 — one global optimum

An Ornstein–Uhlenbeck process

$$dX(t) = \alpha(\theta - X(t)) dt + \sigma dW(t),$$

with the root drawn from its stationary distribution about θ . Being stationary, the mean is θ at every tip and the covariance depends on the tips only through patristic distance.

$$\mathbf{y} \sim \mathcal{N}_n(\theta \mathbf{1}_n, s^2 \mathbf{K}(\alpha) + \tau^2 \mathbf{I})$$

Free parameters: θ (the optimum), α (selection strength), s (stationary spread), τ . The scale is the *stationary* standard deviation $s^2 = \sigma^2/(2\alpha)$, which is comparable across values of α ; recover $\sigma = s\sqrt{2\alpha}$.

OUx — one optimum per regime

The same process, but the optimum is piecewise constant along the tree, $\theta(u) = \theta_{r(u)}$. Only the mean changes.

$$\mathbf{y} \sim \mathcal{N}_n\left(\mathbf{W}(\alpha) \boldsymbol{\theta}, s^2 \mathbf{K}(\alpha) + \tau^2 \mathbf{I}\right), \quad \boldsymbol{\theta} \in \mathbb{R}^K$$

$\mathbf{W}(\alpha)$ is the **exposure matrix**. Let $\mathcal{P}(i)$ decompose tip i 's root-to-tip path into maximal segments (k, a, b) : regime k occupying the time interval $[a, b]$. Then

$$\mathbb{E}[y_i] = \underbrace{e^{-\alpha H} \theta_{r(0)}}_{\text{root}} + \sum_{(k, a, b) \in \mathcal{P}(i)} \theta_k \left(e^{-\alpha(H-b)} - e^{-\alpha(H-a)} \right) = [\mathbf{W}(\alpha) \boldsymbol{\theta}]_i.$$

Each row records how much of that tip's ancestry sits in each regime, discounted by how quickly the process forgets. The segment terms telescope to $1 - e^{-\alpha H}$ and the root term supplies the rest, so $\mathbf{W}$ is **row-stochastic**: $\sum_k W_{ik} = 1$, and the tip mean is a genuine weighted average of the optima.

Relationships

- OU1 is OUx with every θ_k equal: an *interior* restriction, nothing degenerates.
- BM1 is the $\alpha \rightarrow 0$ limit — but only after the diverging common term $\sigma^2/(2\alpha)$ is absorbed by the free mean, since $s^2 \mathbf{K}(\alpha) \rightarrow \left[\frac{\sigma^2}{2\alpha} - \sigma^2 H \right] \mathbf{1}\mathbf{1}^\top + \sigma^2 \mathbf{C}$. The approach is numerically singular: $\mathbf{K}(\alpha) \rightarrow \mathbf{1}\mathbf{1}^\top$, which has rank one. With a *fixed* rather than stationary root the limit is clean, $\mathbf{K} \rightarrow \mathbf{C}$, with no divergence.
- Counting free parameters: BM1 has 3, OU1 has 4, OUx has $3 + K$.